



AR to help partially blind

Smart AR glasses that will help the partially blind see objects around them clearly being developed by researchers <http://digit.in/ARBlind>



A costly coding mistake

Ethereum Parity Wallet's "multisignature" wallets contain an error that makes it impossible for users to move their funds out. <http://digit.in/EthOops>

Evolution of Biometrics

T

echnology has evolved in complex and dynamic ways especially when it comes to the manner

in which we communicate with one another. At the heart of this, is a rich and growing field called biometrics that is redefining the way in which we identify and recognise each other. When we talk about biometrics in the modern context, we either refer to technology that we use in our day-to-day lives such as the fingerprint sensor (and now, the iris sensor) on our phones, or a futuristic scenario that involves the face recognition mechanism as seen in the Tom Cruise sci-fi flick, *Minority Report*. But, what exactly does it mean? At its most basic level, biometrics is exactly what it means when broken down - the word 'bio' means life and 'metrics' means measurement. Any biological or physical marker that can be used to establish a person's identity qualifies to be called as a biometric. In fact, the practice of biometrics can be traced to a long time back in history.

HOW THE FRENCH DID IT

In the late 1800s, Alphonse Bertillon, a French police officer was said to have been frustrated in the manner in which data of captured criminals was recorded by the police. He came up with a method called Anthropometrique, which was a set of body measurements that reportedly had a failure rate of only one in 286,435,456. Law enforcement agencies around the world were quick to adopt this system, until its flaws were soon discovered. Because of the inaccuracy of the instruments at the time,

measurements of the same person tended to differ based on the person taking the measurements. Moreover, there were instances wherein two or more people could have similar or almost identical measurements. In response, Alphonse Bertillon went on to use and pioneer the more effective parallel biometric system - the fingerprint identification system. In fact, Bertillon is also credited to have explored other biometric identifica-

tion methods such as the iris recognition at the time, but only in theory.

THE FINGER PRINT

It is rather difficult to pinpoint when exactly the fingerprint identification system came to be as the standard in the world of biometrics. A reason for this is said to be the difference in timeline of this system's use as a unique anthropomorphic marker and as a means of identification, especially in law enforcement and governance around the world.

In 1880, Dr Henry Faulds published a paper in *Nature* talking about the use of fingerprints as a means of identification of criminals. It was during his visit to Tokyo, where he is said to have noticed fingerprints on pottery, that prompted him to think that something similar could be used for authentication and identification. In the late 1870s, William Herschel in colonial India established the use of fingerprints (handprints, in this case) on government documents to prevent forgery. In the 1890, Sir Francis Galton, who was well-versed with Faulds' research, also made significant progress in this field by establishing the ridge as a unique characteristic in the fingerprint.



The Bertillon Anthropometric system